CARNEGIE MELLON UNIVERSITY
Heinz College of Information Systems and Public Policy

Donor Matching Tool for Nonprofits

FINAL REPORT

Capstone Project with KM Strategies Group

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1. Executive Summary

Small nonprofits in Pennsylvania often face significant challenges in identifying suitable foundation funders. Limited staff capacity, lack of access to expensive prospecting tools, and the time-intensive nature of analyzing grant histories make donor discovery inefficient and often ineffective. As a result, many organizations rely on intuition, broad outreach, or incomplete information when pursuing institutional funding.

To address this problem, this project develops a prototype donor matching tool that leverages publicly available IRS Form 990 data to generate ranked funder recommendations. The system integrates cleaned grant-level data and aggregated funder profiles, and applies a structured matching logic based on three key signals: thematic relevance, geographic alignment, and grant-size fit. In addition to producing ranked shortlists, the prototype provides transparent explanations and supporting grant evidence, enabling users to understand and evaluate each recommendation. However, the tool is designed as a decision-support system rather than a predictive model. It does not estimate funding probability and is currently limited to Pennsylvania-based foundations with annual giving above $1 million.

The data analysis reveals several important insights that directly inform system design. First, grantmaking activity is highly concentrated, with a small number of large foundations accounting for the majority of total funding, while many smaller foundations operate with fewer but often larger individual grants. Second, geographic alignment plays a critical role, as a majority of foundations direct most of their funding within Pennsylvania, making locality a strong signal for matching. Third, data limitations—particularly missing mission statements and inconsistent purpose labeling—necessitate the use of structured classification systems such as NTEE codes to ensure reliable recommendations.

Looking forward, the recommended path includes expanding geographic coverage beyond Pennsylvania, improving thematic classification through enriched data sources, and integrating additional performance-related signals such as grant outcomes or nonprofit characteristics. Further development should also include user testing with nonprofit stakeholders to evaluate usability, trust, and real-world effectiveness. With these improvements, the tool has the potential to become a scalable and accessible solution that significantly reduces the cost and effort of donor prospecting for small nonprofits.

2. Project Background & Problem Framing

This section establishes the organizational and policy environment that motivates the prototype, defines the population of nonprofit users, and traces the project's scope decisions back to documented stakeholder needs. The goal is to make clear why a funder-matching tool is needed in Pennsylvania, who stands to benefit, and what constraints shape the design choices presented in later sections. Small nonprofits often have fewer full-time staff dedicated to fundraising, with limited or no institutional access to paid prospect-research databases. Identifying a single appropriate foundation typically requires extensive research of multiple annual reports and Form 990 filings, cross-referencing past grant recipients, evaluating geographic and mission alignment, and judging whether the foundation's typical grant size matches the nonprofit's needs. For an organization with fewer staff, this work competes directly with program delivery time. The result is that nonprofits often engage in a pattern of high-effort, low-yield prospecting, in which small nonprofits either apply broadly, rely on word-of-mouth referrals, or simply do not pursue institutional funding at all.

The Gap in Existing Tools

Several commercial and nonprofit-sector tools already attempt to address funder discovery. The most widely known are Candid's Foundation Directory, Instrumentl, and GrantStation. These tools are valuable and, for well-resourced development offices, often sufficient. However, two structural gaps remain:

Cost and access: The most comprehensive paid platforms operate on annual subscription models that range from several hundred to several thousand dollars per year, with tiers that include the data fields most useful for prospecting (full grant histories, recipient details, and search functionality across them). The organizations with the greatest need for efficient prospecting are also the ones least likely to access the tools that would provide it.

Explanation and transparency. When existing tools rank or recommend funders, the rationale is rarely surfaced to the user. The user sees the result but not the reasoning, making it difficult to learn from the system or challenge its recommendations.

This project addresses these gaps by building on free, public data and producing ranked shortlists rather than raw search results. Additionally, it surfaces the evidence behind each recommendation so that users can evaluate for themselves.

Why Pennsylvania, and Why the $1M Threshold

The decision to scope the MVP to Pennsylvania foundations giving more than approximately $1 million annually reflects stakeholder direction and three supporting considerations.

Stakeholder direction: The Pennsylvania scope was specified by our project partner, KMSG, whose operations and nonprofit engagement are concentrated in the state. Anchoring the MVP in Pennsylvania, therefore, aligns the prototype directly with the geography where KMSG's clients are most likely to apply it.

Geographic tractability: Pennsylvania has a large and well-documented foundation sector, anchored by major metropolitan areas in Philadelphia and Pittsburgh, and supported by meaningful regional philanthropy in Harrisburg, Lancaster, Allentown, Erie, and Scranton. Restricting the dataset to a single state allows the project to validate its data pipeline, matching logic, and explanation layer.

Data quality at scale: The $1M annual giving threshold is a filter that concentrates the dataset on foundations whose grantmaking is substantial enough to be consistently and completely reported on

Form 990 Schedule I. Smaller foundations file less detailed disclosures and contribute disproportionately to the data-cleaning challenges already documented in the deck.

Stakeholder Map

Tier	**Stakeholder**	**Relationship to the tool**	**Primary interest**
Primary users	Executive Directors of small nonprofits	Direct user	Fast, credible shortlists they can act on
Primary users	Development staff and grant writers	Direct user, prospect research workflow	Efficient discovery, defensible match rationale
Secondary users	Nonprofit boards and finance committees	Indirect — review prospecting outputs	Transparency in how funders were selected
Affected parties	Foundations themselves	Subject of the recommendations	Accurate representation of their giving; appropriate inbound applications
Affected parties	Nonprofits *not* surfaced by the tool	Indirectly affected by visibility decisions	Fair treatment; not being excluded by data gaps

COMPARABLE TOOLS

Tool	Data source	Access model	Core functionality	Explanation of recommendations	Key limitation for small nonprofits
Candid Foundation Directory	Aggregated 990s, foundation websites, surveys	Tiered annual subscription	Filtered search, foundation profiles,	Limited surface filters, not reasoning	Subscription cost; search-engine interaction model
ProPublica Nonprofit Explorer	IRS 990 filings	Free	Search and view filings	None	Built for research, not for matching
Charity Navigator	IRS Form 990, charity self-reported	Free	Charity ratings and search for	Star ratings with breakdowns; does not explain	Oriented toward donor-side

	data		*donors*; star-based scoring on financial health, accountability, and impact	funder-nonprofit fit	evaluation, not nonprofit-side funder prospecting
Instrumental	Aggregated grant data, deadlines, and foundation profiles	Tiered annual subscription	Matching, deadline tracking, workflow tools	Match scores shown; full rationale limited	Subscription cost, oriented toward larger development teams
Raw IRS 990 filings	IRS public data	Free	Full primary-source disclosures	None - user must interpret	Requires substantial time to use

3. Project Scope, Deliverables & Success Criteria

This section formally documents the boundaries of what the project will and will not attempt, the deliverables that constitute the capstone submission, and the measurable criteria against which those deliverables will be evaluated. These commitments were established early in the project and have been refined through iterative conversations with the KMSG project partner and faculty advisors.

3.1 Scope Boundaries

In-Scope:

The following activities and artifacts fall within the project boundary:

Data acquisition and pipeline: The project ingests IRS Form 990 public XML data, covering Pennsylvania-based private foundations with annual charitable giving exceeding approximately $1 million. A custom Python-based ETL pipeline was built to parse thousands of federal tax archive files, extract relevant grant-level records, and convert raw XML into a structured, queryable dataset.

Data cleaning and enrichment: Substantial effort was invested in addressing known quality issues in the raw data. These include funder name and EIN mismatches (where 9,164 unique EINs mapped to only 7,638 unique names due to typographical variations, tax-preparer name instead of funder name), inconsistent location strings, negative, zero, and outlier values in grant amount and revenue fields, and the consolidation of over 40,000 unique free-text purpose descriptions into a manageable set of purpose categories.

Analytical outputs: The project produces ranked funder lists generated through a matching mechanism that combines TF-IDF cosine similarity on grant purpose text, geographic fit scoring, and grant-amount alignment scoring. Results are accompanied by evidence citations, the specific text-similarity score, the geographic-match rationale, the amount-fit score, and the extracted keywords, which explain each recommendation. A retrieval-augmented generation (RAG) layer sits on top of this matching pipeline, allowing the system to synthesize its structured scoring outputs into natural-language explanations grounded in the underlying data.

Ethics framework and responsible-use guardrails: The project includes a dedicated ethics and governance component that documents intended-use boundaries and identifies key ethical risks (e.g., visibility bias, data limitations, algorithmic influence on philanthropic attention).

Matching logic documentation: The weighting scheme, scoring functions, and filtering thresholds used in the matching mechanism are fully documented, including the rationale for default weights.

Out-of-Scope:

Outreach automation. The tool does not auto-draft donor emails, grant letters, letters of inquiry, or any other outreach materials. Its output is a ranked shortlist with evidence.

Foundations outside Pennsylvania or below the $1M annual giving threshold: As discussed in the project scope, the geographic and size filters are MVP-scoping decisions aligned with KMSG's operational focus. The data pipeline is architected to accommodate expansion to additional states and lower thresholds, but such expansion is not attempted in this iteration.

Predicting award probability: The matching scores produced by the tool represent alignment between a nonprofit's stated needs and a funder's documented giving history. They do not predict the probability that an application will be funded.

Replacing human due diligence: The tool is designed to accelerate the discovery phase of funder prospecting, not to substitute for the relationship-building, proposal development, and judgment that characterize effective fundraising.

3.2 Deliverables

Cleaned Pennsylvania Foundation Knowledge Base. A structured dataset of Pennsylvania foundations meeting the $1M annual giving threshold, derived from IRS Form 990 Schedule I filings for tax years 2020–2023. Each funder profile includes: funder EIN and canonical display name, location (city and state, with ZIP codes where available), revenue, number of grants, grant amount distribution (median, 75th percentile, and maximum), aggregated purpose text, and top recipient states.

Three Documented Query Templates with Field Mappings and Example Outputs. The project defines a set of structured query types that the matching mechanism supports, each with documented field mappings that connect natural-language user inputs to dataset columns. These include: recommending funders by cause area, geography, and grant size; discovering funders through similar-organization analysis; and finding similar organizations, then finding funders.

Exploratory Data Analysis Report. A set of charts and diagnostic analyses documenting the composition, coverage, and quality of the cleaned dataset.

Prototype AI Query Interface (Backend Logic Only). The matching mechanism is implemented as a Python-based backend that accepts structured queries and returns ranked lists of funders with supporting evidence. The mechanism combines three scoring dimensions, including text similarity, geographic fit, and amount fit (comparing the user's requested grant size against the funder's historical grant distribution). These dimensions are combined using configurable weights.

Ethics & Governance Framework. This addresses intended-use boundaries, responsible AI principles applied to design, and key ethical risks identified. This deliverable is discussed in detail in Section 5 of this report.

Final Report. This document. The final report consolidates the project's motivation, context, scope, methodology, findings, ethical analysis, and recommendations for future work into a single comprehensive narrative.

3.3 Success Criteria

Criterion 1: Defensible recommendations across multiple test profiles. The query templates and matching mechanism must return evidence-cited funder recommendations for at least three distinct test nonprofit profiles. "Defensible" means that each recommendation is accompanied by specific, verifiable evidence

Criterion 2: Data completeness diagnostics. The exploratory data analysis (D3) must clearly document where the dataset has gaps—missing fields, sparse categories, geographic blind spots, or temporal coverage limits and articulate how those gaps affect the reliability of recommendations derived from the data.

Criterion 3: Ethics review. This criterion acknowledges that ethical analysis involves judgment and that reasonable reviewers may identify areas for improvement; the standard is that the framework demonstrates serious engagement with the project's ethical dimensions rather than perfection.

Criterion 4: Reproducibility. All outputs must be reproducible from publicly available IRS 990 data using documented steps. This criterion ensures that the project's claims are verifiable and that the tool could, in principle, be maintained or extended by a future team without relying on undocumented processes or proprietary inputs.

4. Data Collection, Pipeline & Cleaning

4.1 Data Sources

The project dataset was built from multiple public and project-specific data sources to create a structured view of nonprofit funding relationships. Rather than relying on a single source, the pipeline combined organization-level nonprofit records, IRS filing data, grant-level XML extraction, and supplemental reference data to improve completeness and usability.

The primary data sources were:

- **IRS Exempt Organizations Business Master File (EO BMF)**

The IRS EO BMF was used as the starting population of nonprofit organizations. For this project, the pipeline focused on Pennsylvania organizations and filtered for 501(c)(3) entities. The BMF provided core organization-level fields including EIN, organization name, address, city, state, ZIP code, subsection code, revenue, income, assets, and NTEE classification. These fields were used to identify candidate funders and establish a baseline organizational profile.

- **IRS Form 990 XML filings**
  IRS Form 990 XML filings were the primary source for detailed filing-level and grant-level information. The pipeline used IRS XML index files for tax years 2020–2024 to identify filing object IDs associated with qualified organizations. These object IDs were then used to retrieve public XML filings and extract Schedule I grant records, including recipient name, recipient EIN, recipient location, grant amount, and purpose of grant. The XML filings were also used to recover organization-level metadata such as mission statements, website addresses, and mailing addresses when available.

- **ProPublica Nonprofit API**
  The ProPublica Nonprofit API was used to enrich organization and filing data for qualified funders. This source provided structured organization information and filing-level financial metrics, including total revenue, functional expenses, and end-of-year assets. ProPublica data was especially useful for creating funder-level tables and linking organizational characteristics to each grant record.

- **Pennsylvania nonprofit reference dataset**
  A cleaned Pennsylvania nonprofit reference file was used as an enrichment and validation source. This dataset helped improve completeness for both funders and recipients by adding or confirming fields such as nonprofit name, EIN, street address, city, state, ZIP code, cause area, NTEE code, assets, and income. The reference dataset was used after IRS and ProPublica extraction to fill missing values and create more usable organization profiles.

Key identifiers and matching fields.

The EIN was treated as the strongest organization identifier and was used as the primary matching key wherever available. When EIN-based matching was not possible, the pipeline used cleaned organization names as a controlled fallback. ZIP codes, organization names, and NTEE codes were standardized to support consistent merging, quality checks, and later analysis.

4.2 Extraction & Enrichment Pipeline

The pipeline was designed as a reproducible, multi-stage process that converts fragmented public nonprofit data into a cleaned grant-level dataset. The final output is structured so that each row represents one grant relationship, with attached funder information, recipient information, grant amount, grant purpose, tax year, and filing context.

4.2.1 Initial organization identification.

The pipeline began by loading the IRS EO BMF dataset for Pennsylvania and filtering to 501(c)(3) organizations. This created the initial population of eligible nonprofit organizations. Organizations with reported revenue of at least $1 million were treated as confirmed high-capacity funder candidates. Organizations with missing revenue were not immediately discarded; instead, the pipeline checked whether they had electronically filed IRS XML records that could be used for revenue validation.

4.2.2 Revenue validation and qualified funder selection.

To avoid excluding organizations solely because of missing BMF revenue data, the pipeline loaded IRS XML index files for tax years 2020–2024 and identified missing-revenue organizations with available filings. For these organizations, the pipeline fetched the most recent XML filing and attempted to extract total revenue directly from the filing. Organizations meeting the $1 million threshold were added to the final qualified funder list. This produced a more defensible funder universe than relying only on the BMF revenue field.

4.2.3 ProPublica organization and filing enrichment

After identifying qualified funders, the pipeline used the ProPublica Nonprofit API to retrieve organization-level and filing-level data for each funder EIN. The resulting tables included funder name, city, state, ZIP code, NTEE code, subsection code, total revenue, total functional expenses, and end-of-year assets. These fields provided structured financial and organizational context that could later be attached to individual grant records.

4.2.4 IRS XML filing retrieval and metadata extraction

The pipeline then linked qualified funders to IRS filing object IDs using IRS XML index files. These object IDs allowed the pipeline to retrieve specific public XML filings. From those filings, the pipeline extracted organization-level fields that are not consistently available in structured datasets, including address, mission statement, and website. When XML-derived address information was missing, the pipeline used BMF address fields as a fallback.

4.2.5 Schedule I grant extraction

The core grant dataset was created by parsing Schedule I from IRS Form 990 XML filings. For each available filing, the pipeline extracted donation-level records, including funder EIN, tax year, object ID, recipient name, recipient EIN, recipient city, recipient state, recipient ZIP code, cash grant amount, and grant purpose. The pipeline saved checkpoint files during this process so that long extraction runs could be resumed if interrupted.

4.2.6 Funder metadata attachment

Once Schedule I grant records were extracted, the pipeline merged enriched funder information back onto each grant. This created a more complete grant-level table that included both the transaction itself and the organizational characteristics of the funder. The resulting intermediate dataset included funder identity, funder location, mission, website, NTEE code, financial metrics, recipient information, grant amount, and grant purpose.

4.2.7 Pennsylvania reference data enrichment

The pipeline then used the Pennsylvania nonprofit reference dataset to improve completeness and consistency. This stage matched both funders and recipients to the reference file using EIN first and cleaned organization name second. EIN matching was prioritized because it is more reliable; name matching was used only as a fallback. The enrichment process produced side-by-side original and reference fields, allowing the team to inspect where reference data could fill gaps without overwriting valid original values.

4.2.8 Final cleaning and controlled gap-filling

The final cleaning stage created a cleaned master dataset by preserving original values when available and filling missing values only when a valid enriched value existed. The pipeline also used grouped organization logic to fill stable fields across repeated organizations. For example, if one row for an organization had a ZIP code or cause area but another row was missing it, the pipeline could recover that information from the repeated organization record. This improved completeness while maintaining conservative data handling.

4.2.9 Targeted XML backfill for mission and website fields

Because mission and website data are inconsistently reported, the pipeline included a targeted recovery stage. It identified funders still missing mission or website information, retried XML extraction using available object IDs, and filled only fields that remained missing. This final recovery step improved metadata coverage without changing records that already had valid values.

4.2.10 Final export and exploratory analysis

The final output, final_clean_kmsg_project_df.csv, was exported as the handoff-ready dataset. The pipeline also included exploratory analysis to validate the output and summarize funding patterns, including data completeness, top funders, top recipients, funding by cause area, grant size distribution, and geographic patterns. These checks helped confirm that the final dataset was usable for client-facing analysis and future tool development.

4.2.11 Summary

A detailed, step-by-step explanation of the pipeline, including instructions for running, modifying, and extending the system, is provided in Appendix D: Discovery Pipeline User Guide. This guide is designed for client use and outlines how to reproduce the dataset, adjust parameters such as geography and revenue thresholds, and safely extend the pipeline for future applications.

All code, documentation, and reproducible pipeline materials are available via the project GitHub repository (Appendix F), ensuring transparency, reproducibility, and accessibility for future use and development.

4.3 Cleaning

The primary source table for this project is final_grants_df.csv, a grant-level dataset in which each row represents one historical grant record. The table includes three major types of information: funder attributes, recipient attributes, and grant transaction details. Funder-side variables include identifiers and descriptive fields such as funder_ein, funder_name, funder_address, funder_city, funder_state, funder_zipcode, mission, website, and ntee_code. Financial variables such as totrevenue, totfuncexpns, and totassetsend provide additional information about funder scale. Recipient-side variables include recipient_name, recipient_ein, recipient_city, recipient_state, and recipient_zipcode. Each grant record also contains tax_year, amount, and purpose.

Although this source table is rich, it is not immediately suitable for direct use in a recommendation system or public-facing web application. The raw data mixes multiple levels of information within a single table, contains textual inconsistencies across names and geographic fields, includes free-text purpose descriptions with substantial heterogeneity, and stores several financial and geographic variables in formats that are not yet ideal for downstream processing. Accordingly, the goal of the cleaning pipeline was to transform the raw grant records into a more stable and application-ready data structure.

The data preparation process was designed with two practical objectives in mind. First, the cleaned data needed to preserve record-level detail so that the system could surface concrete historical grant examples as supporting evidence. Second, the data needed to support efficient recommendation and explanation generation at the funder level. To achieve these goals, the pipeline produces two downstream outputs: a cleaned grant-level table and an aggregated funder-level profile table.

4.3.1 Field Standardization and Normalization

The first stage of the pipeline standardizes core text, location, URL, and numeric fields. Text normalization is intentionally conservative. Rather than aggressively merging entities or rewriting names, the cleaning process focuses on removing obvious formatting inconsistency while preserving the semantic content of the source data. A generic text-cleaning function trims leading and trailing whitespace and compresses repeated internal whitespace into single spaces. This logic is applied to major descriptive fields, including funder name, funder address, mission, website, NTEE code, recipient name, and grant purpose.

Geographic fields are then standardized to improve consistency for downstream aggregation and matching. City names are converted into a normalized title-case format, while state abbreviations are standardized to uppercase. ZIP codes are converted into five-character strings wherever possible. This is particularly important because ZIP-related fields may otherwise appear as numeric values, floating-point values, or strings with inconsistent formatting. Converting them into a stable text representation makes them more suitable for later display, filtering, and possible geographic enrichment.

Website URLs are also normalized. Missing or placeholder values such as “N/A” are treated as empty, and URLs without an explicit protocol prefix are standardized by prepending https://. This ensures that valid funder websites can be rendered consistently as clickable links in the application interface.

Finally, key numeric fields are coerced into numeric form using error-tolerant parsing. This includes the grant amount as well as funder-level financial variables such as total revenue, total functional expenses, and ending assets. Standardized numeric fields are essential for later aggregation, outlier flagging, and recommendation scoring.

4.3.2 Outlier Flagging for Grant and Financial Variables

Because philanthropic and nonprofit data are often highly skewed, the cleaning pipeline does not remove extreme values outright. Instead, it flags them for interpretive use while preserving the original values. This choice reflects the fact that unusually large grants or very large organizational finances may represent genuine institutional scale rather than data error.

For grant amounts, outlier detection is performed by tax year. Within each year, the interquartile range (IQR) is calculated and used to define lower and upper bounds. Records with grant amounts outside these bounds are assigned an amount_outlier_flag. A year-specific approach is preferable here because the distribution of grant sizes may vary over time, and a global threshold could misclassify valid observations in years with structurally different distributions.

For funder revenue, a similar IQR-based outlier logic is applied globally, generating a revenue_outlier_flag. Revenue data are sparser and conceptually different from transaction-level grant amounts, so the revenue flag is treated as a descriptive signal. In both cases, the guiding principle is to preserve the full range of real-world observations while making unusually large values visible to downstream systems and analyses.

4.3.3 Rule-Based Purpose Categorization

One of the most important challenges in the dataset is the grant purpose field. The original purpose column contains free-text descriptions that vary substantially in length, specificity, and format. Some entries are highly informative, while others are generic, repetitive, or too broad to support direct thematic comparison. For example, purpose descriptions may range from highly specific program references to generic labels such as "general support," "donation," or "contribution."

To make this field more usable for both analysis and recommendation, the cleaning pipeline implements a lightweight rule-based purpose taxonomy. The taxonomy groups purpose descriptions into interpretable thematic categories, including Education, Health, Human Services, Arts & Culture, Environment, Religion, Animals, Research, General Support, Matching Gift Program, Fundraising / Campaign, Generic / Unspecified, Other, and Unknown.

The classification function uses both the cleaned purpose text and the cleaned recipient name. This design choice improves robustness because many purpose descriptions are too generic to classify accurately in isolation. Recipient names often carry meaningful contextual signals, for example, "University," "Hospital," "Food Bank," or "Church", that can help recover the likely issue area when the purpose text alone is ambiguous. Each record is assigned a purpose_category, which then becomes an important input to funder-level aggregation and system explanation.

This rule-based approach is intentionally simple and interpretable. Although it does not fully solve the problem of semantic ambiguity, it provides a transparent and maintainable thematic layer that supports both exploratory analysis and product functionality.

4.3.4 Construction of the Cleaned Grant-Level Table

After field standardization, numeric coercion, outlier flagging, and purpose categorization, the pipeline outputs a cleaned grant-level table: final_grants_clean.csv. This table preserves the original row-level structure of the source data, so that each row still corresponds to one grant record. However, it adds a set of standardized fields that make the data substantially more usable for downstream application logic.

The cleaned table includes normalized versions of funder, recipient, location, and purpose fields, along with numeric grant and finance fields and the outlier flags described above. The resulting table serves as the project's evidence layer. It retains the specific historical records needed to display sample past grants, supports debugging and auditability, and preserves traceability back to real grant observations.

4.3.5 Construction of the Funder-Level Profile Table

The second major output of the pipeline is funder_profile_final.csv, an aggregated funder-level profile table. This table transforms the original transaction-oriented dataset into a funder-oriented representation in which each row summarizes one funder's historical giving behavior and institutional characteristics.

The aggregation is performed by funder_ein, which serves as the stable funder identifier. For each funder, the pipeline selects the first non-empty standardized value for identifying and descriptive fields such as funder name, address, city, state, ZIP code, website, mission, and NTEE code. These fields define the basic identity and descriptive context of the funder.

The pipeline then calculates a set of funder-level historical summary statistics. These include the minimum and maximum observed years, the number of distinct years active, the number of grant records, total grant amount, median grant size, 75th percentile grant size, and maximum grant size. Median values are also calculated for revenue, expenses, and assets where available. Together, these fields provide a compact summary of organizational scale and grant-making behavior.

To capture geographic tendencies, the pipeline also aggregates recipient-side location information. For each funder, it identifies the most common recipient states and cities and stores them in compact string fields such as top_states_str and top_cities_str. These fields are useful both analytically and operationally, since recommendation logic can use them to estimate geographic fit with a nonprofit user's stated location.

Thematic history is represented through the aggregation of purpose_category into a ranked summary field, top_purpose_categories. In addition, the pipeline generates sample recipient and sample purpose strings, which provide lightweight qualitative summaries of a funder's past activity.

Finally, the pipeline constructs a funder-level text corpus by concatenating the funder mission, cleaned purpose texts, and cleaned recipient names across the funder's historical records. This text_corpus is designed to support textual similarity matching in the recommendation system. Rather than relying on a single short purpose field, the corpus captures multiple layers of information about the funder's stated mission, issue-area history, and recipient portfolio. A Pennsylvania-specific share metric, share_in_pa, is also computed to summarize the extent to which a funder's historical grant-making is concentrated in Pennsylvania.

The resulting profile table is therefore a serving-oriented representation of each funder that can be used directly by the application's matching logic and explanatory interface.

4.3.6 Resulting Data Outputs

The final outcome of the data preparation pipeline is a two-table structure designed to support both analytical transparency and product functionality.

The first output, final_grants_clean.csv, preserves cleaned record-level grant evidence. It is particularly useful for displaying concrete historical grant examples, for validating recommendations against actual past giving behavior, and for maintaining traceability back to the source data.

The second output, funder_profile_final.csv, provides a compact and system-ready representation of each funder. It contains the main fields needed for recommendation, including institutional descriptors, grant-size statistics, geographic summaries, thematic summaries, and a textual profile derived from historical records.

Together, these outputs convert a raw transaction-oriented source table into a more structured data foundation for the website. The cleaned grant-level table supports evidence and transparency, while the funder-level profile table supports efficient recommendation, ranking, and explanation generation.

5. Exploratory Data Analysis

The EDA characterizes the corpus the RAG system will retrieve from and identifies data properties that constrain retrieval design. Each subsection surfaces one finding and states the design choice it implies. Supporting figures and tables are provided in Appendix A.

5.1 Dataset Overview and Completeness

The corpus contains 190,778 grant records from 1,013 Pennsylvania-based foundations (tax years 2020–2024), disbursed to 63,052 unique recipients, totaling $34.9B in grant volume. Record volume grew from 28,757 in 2020 to 55,620 in 2023 due to IRS filing lag rather than rising grant-making activity; 2024 filings are still being processed and should be treated as partial (see Appendix A).

Metric	Grants Dataset	Funder Profile Dataset
Total records	190,778	1,013
Unique foundations	1,013	1,013
Unique recipients	63,052	—
Tax years	2020 – 2024	2020 – 2024
Total grant dollars	$34.9B	$34.9B
Median grant	$18,000	$40,628
Mean grant	$186,037	$922,779

Table 1 Dataset at a Glance

Mean and median diverge sharply at both the grant level ($186K vs $18K) and the funder level ($923K vs $41K), signaling a heavy right-tail distribution that we revisit in §5.4.

Missing data is concentrated in unstructured fields. Websites are missing in 54.4% of profiles, Mission statements are absent in 48% of funder profiles. In contrast, NTEE codes (a standardized IRS classification) are 88% complete, and other structured fields (purpose samples, top-state/city strings, financial percentiles) are >96% complete.

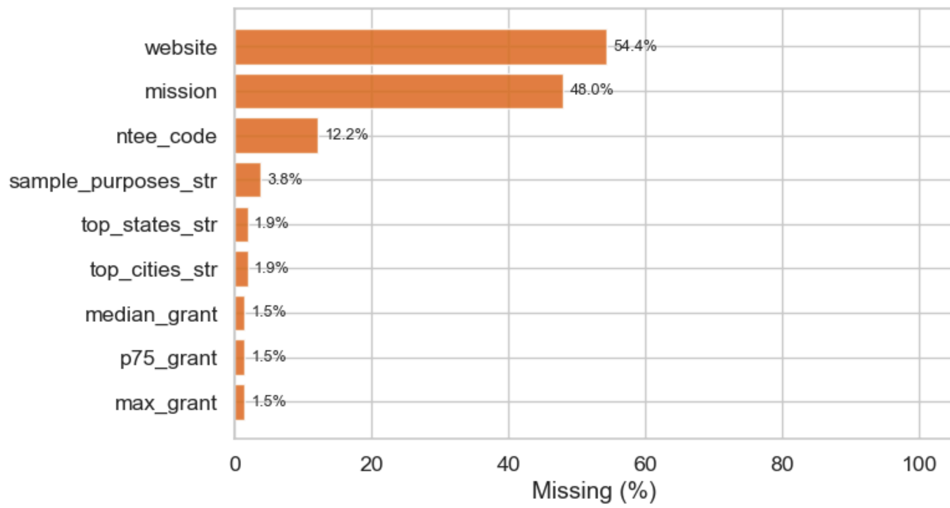

Figure 1 Funder Profile - Missing Data %

Design implication: free-text mission embeddings alone cannot carry retrieval for half the corpus. NTEE codes must serve as the structural fallback when mission text is sparse.

5.2 Geographic Coverage

All 1,013 funders are PA-based, but recipients span the country. Grant activity concentrates in two metro corridors — Philadelphia (ZIPs 19103, 19107, 19104) and Pittsburgh (15213, 15222) — with the top 20 recipient ZIPs capturing a disproportionate share of records (Appendix A). More importantly, at the funder level, 68% of foundations direct ≥90% of their grant dollars to PA recipients.

Design implication: share_in_pa is a high-precision filter for users seeking genuinely PA-local funders and should be exposed as a ranking feature.

5.3 Foundation-Level Characteristics

Foundations vary by two orders of magnitude in activity. Median grants per foundation is 6 while the mean is 188, confirming that a small number of very active funders inflate aggregates. Binning foundations by grant count into Micro (1-5), Small (6-30), Medium (31-100), and Large (100+) tiers, nearly half (48%) are Micro — likely small family or community funds with narrow focus. 58% of foundations give exclusively within PA.

Counter-intuitive finding — smaller foundations write larger individual checks. Micro funders have a median grant of $120K; Large funders have a median of just $16K. The relationship is monotonic across all four tiers.

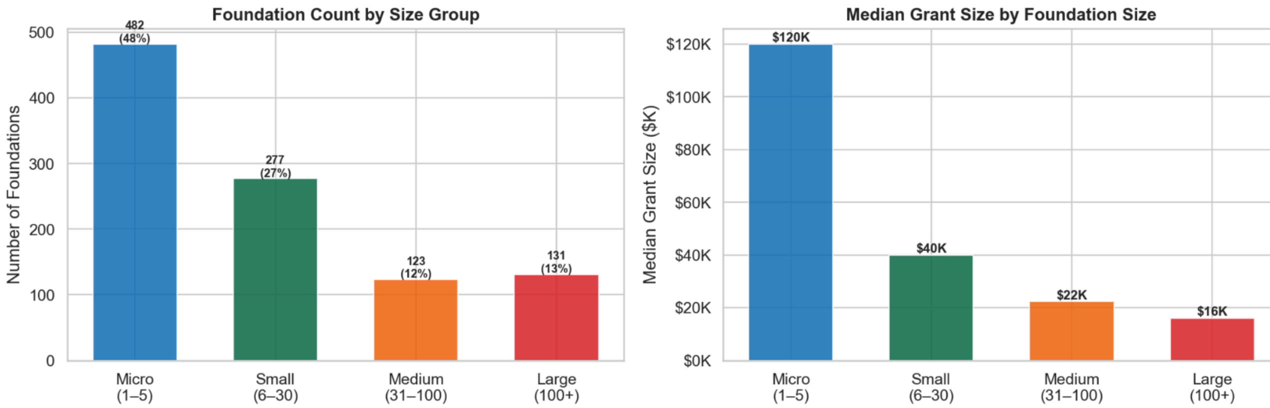

Figure 2 Foundation Size Distribution & Typical Grant Size

Design implication: use **median grant size** (not mean, not total giving) as the primary amount-matching signal. Nonprofits seeking large single awards should be preferentially routed to Micro/Small funders — the opposite of what a naïve "biggest funder" heuristic would suggest.

Size Group	# Foundations	Avg # Grants	Median Grant	Median Assets
Micro (1–5)	482	2.7	$120,000	$13.7M
Small (6–30)	277	12.2	$40,000	$14.6M
Medium (31–100)	123	56.4	$22,302	$11.5M
Large (100+)	131	1367.7	$16,000	$34.1M

Table 2 Profile by Foundation Size Group

Design implication: Use **median grant size** (not mean, not total giving) as the primary amount-matching signal. Nonprofits seeking large single awards should be preferentially routed to Micro/Small funders — the opposite of what a naïve "biggest funder" heuristic would suggest.

5.4 Grant Concentration

Giving is extreme right-tail: the top 1% of foundations (10 organizations) control 75.8% of all grant dollars, and the top 10% control 92.7%.

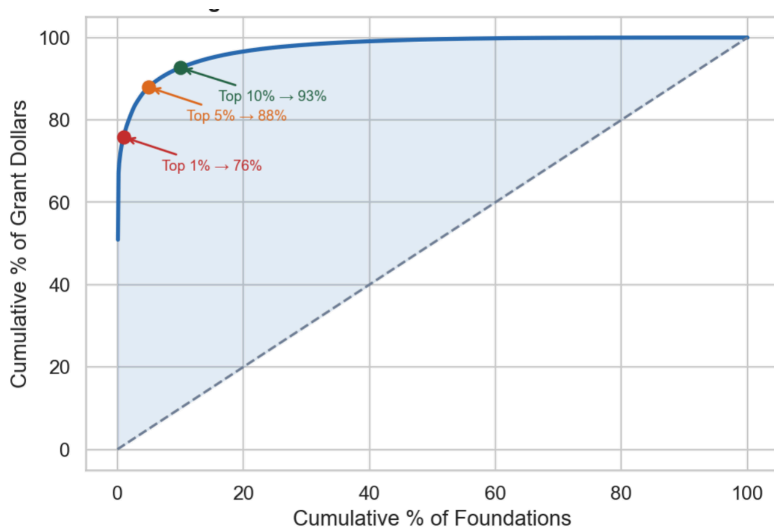

Figure 3 Lorenz Curve - Funder Concentration

Two donor-advised funds — National Philanthropic Trust ($17.8B) and Vanguard Charitable ($5.7B) — alone account for roughly two-thirds of the entire $34.9B corpus. Unlike institutional foundations, DAFs are pass-through vehicles: grant decisions reflect thousands of individual donors, not a coherent organizational mission.

Design implication: DAFs must be tagged (is_daf=True) and handled through a separate matching path, or excluded from mission-based retrieval entirely. Without this separation, the top two funders would dominate every ranking despite having no institutional mission to match against.

5.5 Purpose Category Distribution

"Other" is the dominant category by both count (44%) and dollar volume ($16.2B, 43% of total), indicating significant noise in purpose labeling. Among well-defined categories, Education leads at $8.8B, followed by Health ($4.2B) and Arts & Culture ($1.9B). Research grants are fewest in number but have the highest median size ($32K), suggesting Research funders write fewer, larger checks.

Grant-level purpose labels are unevenly distributed. "Other" is the largest category by both count (44%) and dollar volume (43%, $16.2B), indicating substantial noise in purpose labeling. Among well-defined categories, Education dominates at $8.8B, followed by Health ($4.2B) and Arts & Culture ($1.9B). Research grants are the fewest in number but have the highest median size ($32K), consistent with a small pool of technical, high-dollar funders.

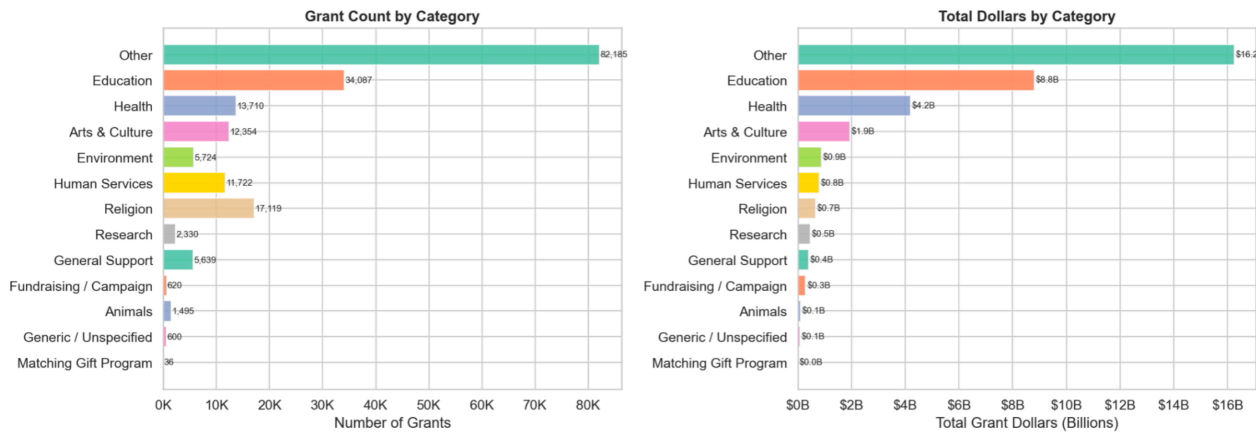

Figure 4 Donation Records by Purpose Category

Design implication: category labels alone cannot drive retrieval when 44% of records are "Other." The pipeline should re-label "Other" grants using their free-text purpose descriptions before indexing, with NTEE codes as a structural backstop.

6. System Design & Architecture

The system was designed as a lightweight, modular pipeline that converts cleaned philanthropic grant data into an interactive, publicly accessible recommendation tool for small nonprofits. At a high level, the architecture consists of four layers: a data layer, a recommendation layer, an application layer, and a presentation layer. Together, these components transform raw historical grant information into ranked funder suggestions, explanation text, and supporting examples that are understandable to nonprofit users.

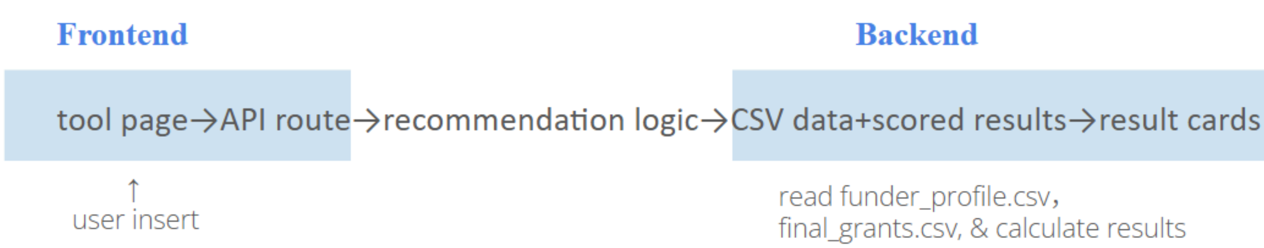

Figure 5. System Architecture

The data layer provides the structured inputs required for the application. The system uses two prepared CSV files generated through a preprocessing notebook: a cleaned grant-level table and an aggregated funder-level profile table. This design reduces runtime complexity and ensures that the application serves standardized, analysis-ready data.

The recommendation layer consumes the processed data and computes funder rankings based on user-entered nonprofit information. It combines thematic similarity, geographic relevance, and grant-size fit to produce a shortlist of candidate funders. In addition to ranking, this layer also supports the generation of explanatory text and the retrieval of representative past grants, allowing the system to provide not only recommendations but also interpretable evidence.

The application layer provides the interface between the frontend and the recommendation logic. A form-based user request is submitted to a server-side API route, where the input is validated, transformed, and passed into the recommendation engine. The resulting funder recommendations are then returned to the client as a structured response object.

Finally, the presentation layer translates this structured output into a nonprofit-facing web experience. It includes a homepage, tool page, explanatory static pages, and reusable user interface components such as result cards, sample-grant displays, and informational sections. This layered architecture was chosen to support both product clarity and maintainability: each major responsibility–data preparation, scoring, request handling, and user presentation–is separated into a distinct part of the system.

6.1 Data Layer

The data layer is built around two downstream CSV files generated from the curated source table final_grants_df.csv: final_grants_clean.csv and funder_profile_final.csv.

The first output, final_grants_clean.csv, preserves a grant-level structure in which each row corresponds to a single historical grant record. Compared with the source table, this file includes standardized text fields, normalized location fields, cleaned URLs, numeric coercions for financial variables, outlier flags, and a rule-based purpose category. This table is designed to preserve historical detail while making the records substantially easier to use in an application setting. In practice, it functions as the system's evidence table: it supports sample past-grant display, record-level traceability, and downstream validation of funder recommendations.

The second output, funder_profile_final.csv, aggregates the cleaned grant-level data into a funder-level profile table in which each row represents a single funder. This table contains the key fields required for recommendation and user-facing display, including funder identity fields, mission, website, NTEE code, grant-size statistics, geographic summaries, thematic summaries, and a text corpus derived from mission statements, grant purposes, and recipient names. This profile table functions as the serving table for the recommendation engine. Instead of requiring the application to recompute funder behavior dynamically from the full transaction table, the aggregated profile table provides a compact and efficient representation of each funder's historical characteristics.

This two-file structure supports both interpretability and performance. The cleaned grant table retains concrete examples of historical giving, while the funder profile table provides the structured summaries needed for ranking and explanation generation.

6.2 Recommendation Layer - RAG

The recommendation layer converts nonprofit user inputs into ranked funder suggestions with interpretable rationales. It is implemented as a Retrieval-Augmented Generation (RAG) pipeline.

Why RAG. RAG is a framework that pairs a retrieval step over a trusted knowledge source with a generation step performed by a language model. Rather than relying on the LLM's parametric memory, the model is conditioned at inference time on evidence pulled directly from our funder dataset. This design offers two practical benefits for our use case. First, it enables semantic retrieval:

queries are matched to funder profiles based on meaning rather than exact keyword overlap, so a nonprofit describing "after-school tutoring for underserved youth" can surface funders whose historical grants describe "K–12 educational equity programs," even without shared vocabulary. Second, because every recommendation is grounded in retrieved records, the generated rationale is verifiable and traceable back to real grant history, rather than fabricated by the model.

Inputs. The layer takes a small set of nonprofit-facing fields entered on the tool page: issue-area keywords, a short mission or context description, city, state, desired grant amount, and optionally annual operating budget. These inputs are validated and normalized before entering the pipeline.

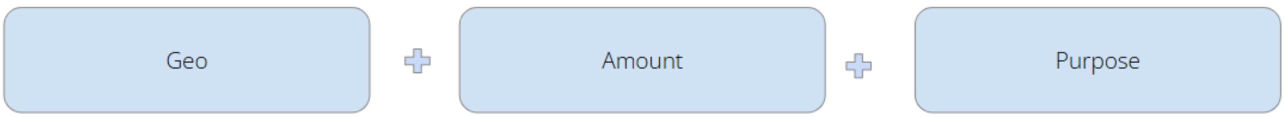

Figure 6. Matching mechanism

Retrieval. Hybrid scoring over funder profiles. Each user query is compared against every funder profile along three signals:

The first is thematic relevance. Each funder profile contains a text corpus built from the funder's mission, grant purpose history, and recipient history — a compact textual representation of the funder's thematic footprint. The user's issue area and mission text are compared against this corpus to produce a topic-alignment score.

The second is geographic relevance. Because nonprofits often operate within a local or regional context, the system checks whether a funder has historically given in the user's geography. Each funder profile stores its most common recipient states and cities, which are compared against the user's stated location.

The third is grant-size fit. Even a thematically aligned funder may not be a realistic match if the requested amount falls well outside its historical giving range. The system compares the user's desired amount against summary statistics from the funder profile, particularly the median and 75th-percentile grant size.

These three signals are combined into a weighted score. Thematic similarity receives the highest default weight, reflecting the intuition that issue-area fit is the strongest basis for initial shortlist generation, while geography and grant-size fit serve as secondary signals that meaningfully refine practical relevance. The top-K funders are returned as the retrieval output.

Augmentation. For each retrieved funder, the engine pulls a small number of representative historical grants from final_grants_clean.csv. These records, together with the funder's profile attributes, form the structured context passed to the generation step.

Generation. The retrieved profile attributes and sample grants are passed to an LLM, which produces a concise rationale explaining why each funder was selected — grounded in theme, geography, and amount fit. Unlike a templated justification, this free-text output conditioned on structured evidence makes the recommendation interpretable and gives the user concrete examples of how the funder has supported similar organizations or purposes in the past.

Cost considerations. Because the generation step relies on an external LLM API, each recommendation request incurs a per-token cost determined by the size of the retrieved context (funder profile attributes plus sample grants) and the length of the generated rationale. To keep operating costs predictable and scalable for a nonprofit-facing tool, the pipeline is designed to limit context size in two ways: by capping the number of retrieved funders (top-K) and the number of sample grants per funder passed into the prompt, and by constraining the generated rationale to a short, bounded length. This keeps per-query token usage low while preserving enough evidence for the model to produce a meaningful explanation. In practice, this design makes the marginal cost per recommendation small enough to support routine use without requiring per-request throttling.

6.3 Application Layer

The application layer is implemented as a web application with a server-side recommendation API. This layer is responsible for receiving user input, validating it, invoking the recommendation engine, and returning the results in a structured format suitable for frontend rendering.

On the client side, the tool page collects nonprofit user inputs through a form-based interface. When the user submits the form, the frontend sends a request to the recommendation API route. This route serves as the operational bridge between the interface and the backend recommendation logic.

The API first validates the submitted input against a predefined input schema. This step ensures that fields such as issue-area keywords, state abbreviation, and requested grant amount are in a usable format before any recommendation logic is applied. Once the request passes validation, the API calls the recommendation module, which loads the prepared data tables, computes funder scores, and returns a ranked list of candidate funders. If recommendations are found, the system also generates explanatory text and attaches sample historical grant records.

The response returned by the API is a structured object containing the ranked results, explanatory information, supporting examples, and metadata about the data source. This structure allows the frontend to render recommendations in a predictable and modular way.

This application-layer separation is important for maintainability. It allows the frontend to remain focused on user experience, while the backend handles validation, ranking, and data access. It also creates a clean path for future extension. For example, if the system later moves from CSV-based serving to a database-backed implementation, the API interface can remain stable while the underlying data access logic changes behind the scenes.

6.4 Web Interface and Product Design

The presentation layer translates the data and recommendation logic into a nonprofit-facing user experience. The website is organized around a small number of clear pages: a homepage, an interactive tool page, and supporting informational pages such as About, Method, Disclaimer, and Contact. This structure was chosen to balance usability, trust, and transparency.

The homepage functions as the product introduction layer. Its role is to clearly communicate the value proposition of the system: helping small nonprofits identify potential funders using public grant history data. The homepage presents a concise explanation of the tool, a simple call to action, and an accessible overview of how the recommendation process works. The homepage also establishes the institutional context of the project, including the collaboration between Carnegie Mellon University's Heinz College and KMSG.

The tool page serves as the core interaction layer. Here, nonprofit users enter a short description of their mission or issue area, their location, and the amount of funding they are seeking. The page then displays ranked funder results in a card-based interface. Each funder card emphasizes the information most relevant to a nonprofit user: the funder's name, website, typical grant size, geographic footprint, topical tendencies, explanation of fit, and sample past grants. This presentation choice is deliberate: Hiding raw tables or technical score outputs, the system packages the recommendation into an interpretable and action-oriented format.

Supporting informational pages play an important role in the overall product architecture. The Method page explains, in plain language, how the tool uses public data and recommendation logic. The Disclaimer page clarifies that the system is a decision-support tool and does not guarantee funding success. The Contact page provides institutional context and collaboration information. Together, these pages strengthen user trust and align the system with the expectations of a public-interest, university-associated project.

In design terms, the interface emphasizes clarity, restraint, and credibility. The site is intentionally simpler than a large nonprofit data portal. Its goal is to guide a user from nonprofit need statement to a plausible shortlist of funders. This product framing is central to the architecture itself: the website is a structured decision-support experience designed to make historical grant data usable for organizations with limited time, staffing, and research capacity.

7. Prototype Outputs & Query Examples

This section demonstrates the prototype's capabilities through a set of concrete, annotated worked examples. Each example reflects a realistic nonprofit query and shows how the system translates user input into ranked funder recommendations with supporting evidence. The outputs are presented in their intended format and annotated to highlight key components, including ranking logic, explanation of fit, and data limitations. Together, these examples allow stakeholders to evaluate the system's relevance, transparency, and practical usefulness in real-world donor prospecting scenarios.

7.1 Worked Example: Template 1

Input:

- Issue area: Youth mentoring, after-school programs
- Location: Pittsburgh, PA (Allegheny County)
- Desired grant amount: $50,000
- Mission/context: A small nonprofit providing after-school mentoring and career readiness programs for underserved high school students in Pittsburgh.

Output:

Rank	Foundation Name	Median Grant Size	Primary Focus	Geographic Fit	Example Grant Evidence	Confidence
1	New Sun Rising	~$59K	Arts, Education, Community	Strong (PA-focused)	$61K grant to local arts org (Pittsburgh, 2021)	High
2	McAuley Ministries	~$50K	Education, Human Services	Strong (PA-heavy)	$50K youth employment program (Pittsburgh, 2020)	High
3	InnovatePGH	~$52K	Education, Innovation	Strong (local)	$59K entrepreneurship program (Pittsburgh, 2023)	Medium
4	Neighborhood Allies	~$15K	Community Development	Strong (PA-focused)	$60K community investment grant (2022)	Medium
5	Jewish Healthcare Foundation	~$68K	Health, Education	Strong (PA-focused)	$50K community education initiative (2020)	Medium

Analysis & Interpretation:

The top-ranked funders show strong alignment across mission, location, and grant size, indicating that the model prioritizes organizations with both thematic relevance and practical funding fit. For example, New Sun Rising and McAuley Ministries closely match the nonprofit's focus on youth and education while operating primarily in Pennsylvania.

Lower-ranked funders reflect partial mismatches. Some have strong geographic presence but weaker alignment in typical grant size or mission focus, which reduces their ranking.

Each recommendation is supported by specific past grant evidence, improving transparency and allowing users to assess credibility. Overall, the results demonstrate that the system produces relevant and interpretable shortlists, while serving as a decision-support tool rather than predicting funding outcomes.

7.2 Worked Example: Template 2

Input

- Issue area: Environmental education, sustainability programs
- Location: Pittsburgh, PA
- Search radius: 50 miles

- Desired grant amount: Not specified
- Mission/context: A nonprofit focused on environmental education initiatives, including community workshops, school partnerships, and sustainability awareness programs in the Pittsburgh region.

Output:

Rank	Foundation Name	Median Grant Size	Primary Focus	Geographic Fit	Example Grant Evidence	Confidence
1	New Sun Rising	~$59K	Education, Community	Strong (PA-focused)	$75K education program (Pittsburgh, 2021)	High
2	Hamot Health Foundation	~$72K	Health, Education	Medium (PA, not local)	$70K university grant (2022)	Medium
3	PA Alliance Foundation	~$31K	Community, Education	Strong (PA-wide)	$75K community program (2022)	Medium
4	Jewish Healthcare Foundation	~$68K	Health, Education	Strong (local)	$74K healthcare initiative (2022)	Medium
5	Passavant Hospital Foundation	~$10K	Health, Community	Strong (local)	$10K community program (2022)	Low

Analysis & Interpretation:
Top-ranked funders reflect strong alignment with the target grant size (~$75K) and a demonstrated focus on education or community-related initiatives. For example, New Sun Rising ranks first due to close alignment in both funding level and Pennsylvania-based activity.

Mid-ranked funders illustrate trade-offs between geography and mission fit. Some, like Hamot Health Foundation, offer strong funding capacity but weaker geographic alignment, while others show broader mission coverage beyond environmental education.

Lower-ranked results highlight weaker financial alignment, where typical grant sizes fall below the requested amount, reducing practical fit despite local presence.

Overall, the output demonstrates that the system balances grant size, thematic relevance, and geographic signals, producing interpretable recommendations supported by concrete grant evidence.

7.3 Worked Example: Template 3

Input:

- NTEE code: B40 (Higher Education)

- Location: Philadelphia, PA
- Desired grant amount: $100,000
- Mission/context: A nonprofit supporting college access and workforce readiness for low-income students in Philadelphia, providing academic advising, career training, and scholarship support.

Output:

Rank	Foundation Name	Median Grant Size	Primary Focus	Geographic Fit	Example Grant Evidence	Confidence
1	Charles E. Kaufman Foundation	~$150K	Higher Education, Research	Strong (PA-focused)	$100K grant to Allegheny College (2020)	High
2	New Sun Rising	~$59K	Education, Community	Medium (PA, not Philly-focused)	$100K workforce-related program (2021)	Medium
3	PA Alliance Foundation	~$31K	Community, Education	Strong (PA-wide)	$108K program support (Philadelphia, 2023)	Medium
4	Greater Philadelphia Community Alliance	~$124K	Human Services	Strong (Philadelphia)	$89K community support grant (2022)	Medium
5	Philadelphia Poverty Action Fund	~$55K	Poverty, Education	Strong (Philadelphia)	$60K grant to Community College of Philadelphia (2023)	High

Analysis & Interpretation:

Top-ranked results are driven by direct alignment with similar organizations (B40 Higher Education). For example, the Charles E. Kaufman Foundation ranks first due to consistent funding of colleges and academic programs in Pennsylvania, making it a strong structural match.

Mid-ranked funders reflect indirect or partial similarity, where support is given to broader community or education-related initiatives rather than strictly higher education institutions. This results in slightly weaker thematic alignment.

Lower-ranked results highlight the importance of geographic and mission trade-offs. While some funders (e.g., Philadelphia-based organizations) show strong local alignment, their funding focus may be broader (e.g., general support or human services), reducing precision.

Overall, this example demonstrates that Template 3 effectively identifies funders through observed funding relationships with similar organizations, providing highly interpretable and evidence-backed recommendations for users with well-defined sector classification.

8. Evaluation & Limitations

This section provides an explicit, honest accounting of what the prototype can reliably answer, what it cannot answer, and why — grounded in current data coverage from Section 5.

8.1 What the Prototype CAN Answer

The prototype is able to support a set of well-defined analytical queries based on structured IRS Form 990 data.

First, the system can identify which Pennsylvania-based foundations in the working sample gave the most in a given year, including segmentation by cause area or geography. This enables users to understand high-level funding patterns within the dataset.

Second, the prototype can retrieve foundations that have historically funded organizations with a specific NTEE code, providing named grant evidence (including recipient, amount, and year). This functionality is particularly important for Template 3, where recommendations are derived from observed funding relationships.

Third, the system supports geographic filtering, allowing users to identify foundations operating within a specified region based on mailing address or historical grant activity. This enables localized donor discovery, which is critical for small nonprofits.

Finally, the prototype generates ranked funder recommendations using a combination of thematic relevance, geographic fit, and grant-size alignment. Each recommendation is accompanied by transparent evidence drawn directly from public 990 records, ensuring that outputs are interpretable and verifiable.

8.2 What the Prototype CANNOT Answer (current constraints)

Despite these capabilities, several limitations constrain the scope and reliability of the system.

Sector-level matching depends on the availability and completeness of grantee EIN and NTEE codes. Gaps in these fields reduce recall for Template 3, as some relevant organizations cannot be reliably linked.

Revenue-based filtering is also incomplete. A significant portion of foundations in the dataset lack usable revenue data, limiting the system's ability to incorporate financial capacity as a ranking signal. As a result, some high-capacity funders may not be surfaced.

In addition, all ranked outputs are generated from a defined working sample rather than the full population of Pennsylvania foundations. This means that "Top 10" results should be interpreted as relative to the sample, not as a comprehensive statewide ranking.

The system also does not predict funding outcomes. It identifies historically plausible matches based on past grant behavior but does not estimate the probability of receiving funding.

Finally, the dataset reflects historical filings, which introduces lag. Foundations that have recently changed their priorities or funding strategies may not be accurately represented until new data becomes available.

8.3 Threats to Validity

Several factors may affect the validity and generalizability of the results.

First, selection bias may arise from the use of a subset of foundations for demonstration purposes. This sample may not fully represent the diversity of the Pennsylvania foundation landscape.

Second, temporal lag in IRS filings means that the dataset reflects past activity rather than current strategy. Foundations that have recently shifted focus may be misclassified.

Third, NTEE codes are self-reported and not independently verified. Misclassification or inconsistency in these codes may affect sector-level matching and thematic interpretation.

9. Ethics & Governance

As this project evolved from a data exercise into a functioning recommendation platform, ethics and governance became core design considerations. Funder Compass is intended to help nonprofit organizations identify potential funding opportunities more efficiently, but recommendation systems do more than organize information. They influence what users see first, where attention is directed, and how limited outreach time is spent. In the nonprofit sector, where staff capacity and fundraising resources are often constrained, those effects matter.

For that reason, the team approached ethics and governance as practical requirements for long-term usefulness, credibility, and adoption. A tool like this must not only generate relevant outputs, but also be understandable, appropriately bounded, and trusted by users.

9.1 Trust is Not Separate From Performance

One of the clearest lessons from this project was that trust and performance are closely linked. Even if recommendations are technically strong, they create limited value if users do not understand them or feel confident acting on them.

For this platform, success depends on five connected factors:.

- Useful recommendations that save users time
- Clear explanations of why results appear
- Honest communication of data limitations
- Human confidence in the outputs
- Repeatable governance for future maintenance

This is especially important for nonprofit users, who often cannot afford to spend time pursuing poor-fit opportunities.

9.2 Responsible Use of Public Data

The platform was intentionally built using public and low-cost data sources, including IRS Form 990 filings, Schedule I grant records, and nonprofit organization reference data. This approach supports transparency, affordability, and long-term scalability.

Strengths of Public Data:

- Publicly accessible and verifiable information
- Historical evidence of prior grantmaking behavior
- Broad organizational coverage
- Lower cost than proprietary fundraising databases

Important Limitations:

- Reporting lag between activity and filing availability
- Uneven documentation quality across organizations
- Community impact is not fully captured in structured filings
- Smaller organizations may appear less visible in available records

For these reasons, public data should be treated as evidence, not as the full story. The system is designed to surface patterns and possibilities, while recognizing that final outreach decisions require additional judgment and current research.

9.3 Intended Use of the System

Funder Compass was developed as a decision-support tool designed to help nonprofit organizations reduce the time and cost associated with early-stage funder research. The platform is intended to surface relevant opportunities, organize historical grantmaking signals, and support more informed outreach planning.

Appropriate uses include identifying potential funders aligned with mission or geography, supporting prospect research, understanding historical giving patterns, and prioritizing outreach opportunities for further review.

The platform is not intended to guarantee funding outcomes, replace professional judgment, rank nonprofit worthiness, determine eligibility with certainty, or fully automate fundraising decisions. Recommendations should be interpreted as informed suggestions that require additional review.

Human judgment remains central to appropriate use. Final decisions about relationship-building, application strategy, and fit should always remain with the user..

9.4 Bias and Fairness Considerations

- The tool is designed to support, not replace, professional fundraising judgment.
- The prototype should not be used to rank or evaluate the 'worthiness' of nonprofits.
- Data should be refreshed at least annually; no organization should rely on results more than 18 months old.
- Query logs must be governed by a documented data-use policy; no individual nonprofit's queries should be visible to others.

9.5 Transparency and Explainability

The team prioritized explainability throughout development. In many AI systems, users lose trust when outputs appear without context. For this reason, recommendations were framed as understandable signals rather than black-box conclusions.

Explainability Features Included:

- Visible matching factors such as mission relevance, geography, and grant patterns
- Plain-language explanations for recommendations

- Clear notice that outputs are based on historical data
- Encouragement to verify current fit directly with funders

This approach helps users understand not only what was recommended, but why it was recommended.

9.6 Governance Foundations for Future Use

Although developed as a capstone prototype, future operational use would benefit from clear ownership and governance processes. Responsible systems need operating rules, especially as they scale.

Recommended Governance Structure:

- Ownership: Define who maintains product and data pipeline
- Refresh Cadence: Schedule updates for filings and records
- Monitoring: Detect stale data or degraded outputs
- Feedback Loops: Collect user insights for improvement
- Documentation: Track methodology and logic changes
- Public Boundaries: Maintain clear disclaimers and user guidance

If the platform expands, these functions would become increasingly important to maintaining reliability.

9.7 Alignment with Responsible AI Principles

The team's approach aligns with widely recognized responsible AI principles, including those promoted by the OECD[1]. These frameworks provided a strong external foundation for trustworthy system design. Using an established framework helped ensure the system reflects broader best practices rather than ad hoc standards.

OECD Principle	Application to Funder Compass
Inclusive Growth & Access	Supports smaller nonprofits with limited research capacity
Fairness	Recognizes structural bias in datasets
Transparency	Explains why recommendations appear
Robustness	Uses repeatable pipelines and structured workflows
Accountability	Encourages clear ownership and stewardship

9.8 Summary

Ethics and governance were treated as core operating requirements rather than optional additions. Throughout the project, the team recognized that the strongest recommendation systems are not only accurate, but also trusted, understandable, and responsibly managed. For a platform intended to

[1] OECD. OECD AI Principles. https://www.oecd.org/en/topics/ai-principles.html

support nonprofit fundraising strategy, long-term value depends on whether users can confidently interpret outputs and apply them appropriately within real decision-making environments.

One safeguard built into the project was the use of ***transparent model inputs***. Wherever possible, key recommendation factors such as mission relevance, geography, and historical giving patterns were made visible to users. This helps clarify how recommendations are generated and reduces the risk that outputs are perceived as unexplained or purely algorithmic conclusions.

A second safeguard was maintaining clear ***awareness of data limitations***. The team consistently acknowledged that nonprofit datasets may contain reporting delays, incomplete records, and uneven levels of organizational visibility. By documenting these constraints, the system encourages users to interpret recommendations with appropriate caution rather than over-relying on incomplete information.

A third safeguard was preserving ***human interpretation of recommendations***. Outputs were intentionally framed as discovery suggestions and research starting points rather than final decisions. This design choice keeps strategic judgment, relationship-building decisions, and outreach prioritization in the hands of nonprofit professionals rather than the system itself.

Taken together, these safeguards reflect the broader philosophy behind Funder Compass: AI should expand capacity, not replace judgment. When paired with clear boundaries, transparent methods, and continued human oversight, systems like this can reduce search friction while preserving user control. In that sense, the project demonstrates a practical and responsible path for applying AI in the nonprofit sector.

*For full transparency and responsible use, this project is accompanied by a dedicated Ethics and Governance Memorandum (Appendix E), which outlines data limitations, appropriate use cases, risks of misinterpretation, and guidelines for responsible deployment.

10. Next Steps & Recommendations

Expand Data Coverage
The current dataset is restricted to Pennsylvania foundations over a four-year window (2020–2023). A natural next step is to broaden the pipeline to ingest IRS Form 990 Schedule I data from additional states, extend the historical time range, and increase the number of foundation profiles indexed. Greater coverage would improve the statistical robustness of match rankings and reduce geographic bias in recommendations surfaced to users.

Integrate Public Enrichment Sources
Future iterations could enrich the underlying dataset by joining it with publicly available contextual data — for example, USPS ZIP centroid coordinates to improve geographic proximity filtering, or Census-derived poverty and demographic indices at the county level. These additions would enable more nuanced matching signals on the website, such as aligning funders with the socioeconomic characteristics of a nonprofit's service area.

Incorporate NTEE Sector Classification
The current model has limited capacity to match on fine-grained cause areas (e.g., animal welfare, arts education) due to inconsistent sector tagging. Integrating a standardized NTEE (National Taxonomy of Exempt Entities) code lookup — for which a viable public source has already been identified — would substantially improve the precision of thematic matching and enable cause-area filtering on the website.

Develop a Governance & Data Stewardship Framework

For any institutional deployment beyond the current prototype, a formal governance document should be established. This would cover data refresh cadence (e.g., annual ingestion of new IRS filings), access and usage policy, version control for the underlying dataset and model logic, and protocols for handling data quality issues or source changes. Given that Kindred has expressed willingness to maintain the platform long-term, this document would serve as a critical handoff artifact.

Build Outreach Automation Features

A high-value extension of the current tool would be outreach assistance — for instance, generating a draft grant inquiry letter or donor outreach email tailored to a specific funder's known giving priorities and historical grants. This would move Funder Compass further along the workflow from discovery to action, directly reducing the time burden on nonprofit staff.

Conduct Systematic User Evaluation with Nonprofit Pilots

The pilot survey initiated toward the end of this project should be formalized into a structured evaluation study with a cohort of Pennsylvania nonprofits. Key metrics to capture include time savings in funder research, satisfaction with match quality, and whether AI-generated explanations are perceived as trustworthy and actionable. This evidence base would be essential for any future funding or scaling decisions.

Nonprofit Outreach & User Validation

Conduct structured outreach to a small cohort of Pennsylvania-based nonprofits to pilot the tool and collect feedback on match quality, usability, and trust in AI-generated results.

Media & Sector Outreach

Engage local media and relevant nonprofit networks (e.g., PANO) to raise awareness of Funder Compass and support broader adoption among the organizations it is designed to serve.

Handoff Documentation

Prepare comprehensive handoff materials — including a product management guide, annotated code repository, data dictionary, and AI use policy — to ensure the platform can be sustainably maintained and extended beyond the capstone project.

11. Conclusion

This project demonstrates that AI-assisted donor matching using public data is both practical and valuable when paired with strong design, transparent methods, and clear user boundaries. Through the partnership between Carnegie Mellon University Heinz College and KM Strategies Group, the team developed Funder Compass, a prototype that transforms complex IRS Form 990 grant records into an accessible recommendation tool for nonprofits seeking funding opportunities. Instead of requiring users to manually search filings, the system generates ranked funder shortlists supported by historical evidence and clear rationale.

The project addressed a meaningful market gap. Many nonprofits lack the budget, staff time, or technical capacity to use commercial prospecting tools or interpret raw public datasets. This prototype shows that public filings can be cleaned, structured, and operationalized into a lower-cost alternative that expands access to smarter donor research.

Several core accomplishments were delivered. The team built a repeatable ETL pipeline, created grant-level and funder-level datasets for system use, designed a multi-factor recommendation engine based on topic, geography, and grant-size fit, and embedded ethics and governance principles directly into the platform. Just as importantly, the analysis surfaced key insights around funding concentration, geographic giving behavior, and the importance of transparent use of imperfect public data.

Overall, the project shows that AI can responsibly expand nonprofit capacity when used as decision support rather than decision replacement. With continued refinement, stewardship, and user testing, Funder Compass has strong potential to become a scalable public-interest resource for the nonprofit sector.

Appendix A — Data Dictionary

Column	Description
funder_profile_final.csv — Identity	
funder_ein	Unique tax ID — primary key
funder_name	Foundation legal name
funder_address	Street address
funder_city	City
funder_state	State (all PA)
funder_zipcode	ZIP code
website	Foundation website URL
mission	Mission statement (free text)
ntee_code	NTEE classification code
funder_profile_final.csv — Grant History (Aggregated)	
min_year / max_year	Active funding period
years_active	Number of years with recorded grants
num_grants	Total grants awarded
total_amount	Cumulative dollars given
median_grant	Median single grant size
p75_grant	75th percentile grant size
max_grant	Largest single grant ever
funder_profile_final.csv — Financial Capacity	
revenue_median	Median annual revenue across years
expense_median	Median annual expenses
assets_median	Median total assets
funder_profile_final.csv — Geographic & Thematic Focus	
top_states_str	Top recipient states with grant counts, e.g. PA(655), NY(21)
top_cities_str	Top recipient cities with grant counts
share_in_pa	Fraction of grants going to PA recipients (0–1)
top_purpose_categories	Top grant categories with counts, e.g. Education(165), Health(103)
funder_profile_final.csv — RAG Text Fields	

sample_recipients_str	Representative recipient org names
sample_purposes_str	Sample grant purpose descriptions
text_corpus	Full concatenated text used for embedding
final_grants_clean.csv — Funder Fields	
funder_ein	Funder tax ID (join key)
funder_name_clean	Cleaned foundation legal name
funder_address_clean	Cleaned street address
funder_city_clean	Cleaned city
funder_state_clean	Cleaned state
funder_zipcode_clean	Cleaned ZIP code
mission_clean	Cleaned mission statement
website_clean	Cleaned website URL
ntee_code_clean	Cleaned NTEE classification code
final_grants_clean.csv — Recipient Fields	
recipient_name_clean	Cleaned recipient org name
recipient_ein	Recipient tax ID
recipient_city_clean	Cleaned recipient city
recipient_state_clean	Cleaned recipient state
recipient_zipcode_clean	Cleaned recipient ZIP code
final_grants_clean.csv — Grant Fields	
amount_num	Grant amount (numeric, cleaned)
purpose_clean	Grant purpose (free text)
purpose_category	Labeled category (Education, Health, Arts & Culture, etc.)
tax_year	Filing year (2020–2024)
amount_outlier_flag	Statistical outlier flag for grant amount
totrevenue_num	Funder total revenue for that filing year
totfuncexpns_num	Funder total functional expenses for that filing year
totassetsend_num	Funder total assets (end of year) for that filing year
revenue_outlier_flag	Financial outlier flag

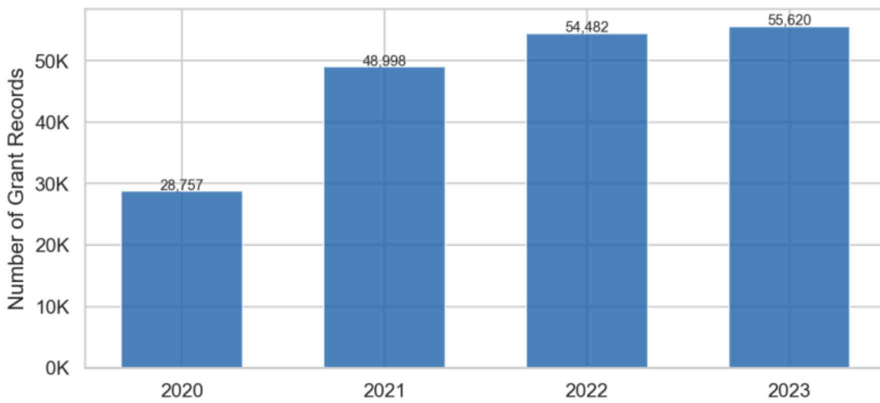

Figure 7 Temporal Coverage (grant records per tax year)

Metric	Value
# Foundations	1,013
Median # grants per foundation	6
Mean # grants per foundation	188
Median total giving	$1.13M
Median single grant size	$40,628
Foundations giving exclusively in PA	588 (58%)

Table 3 Foundation Summary Statistics

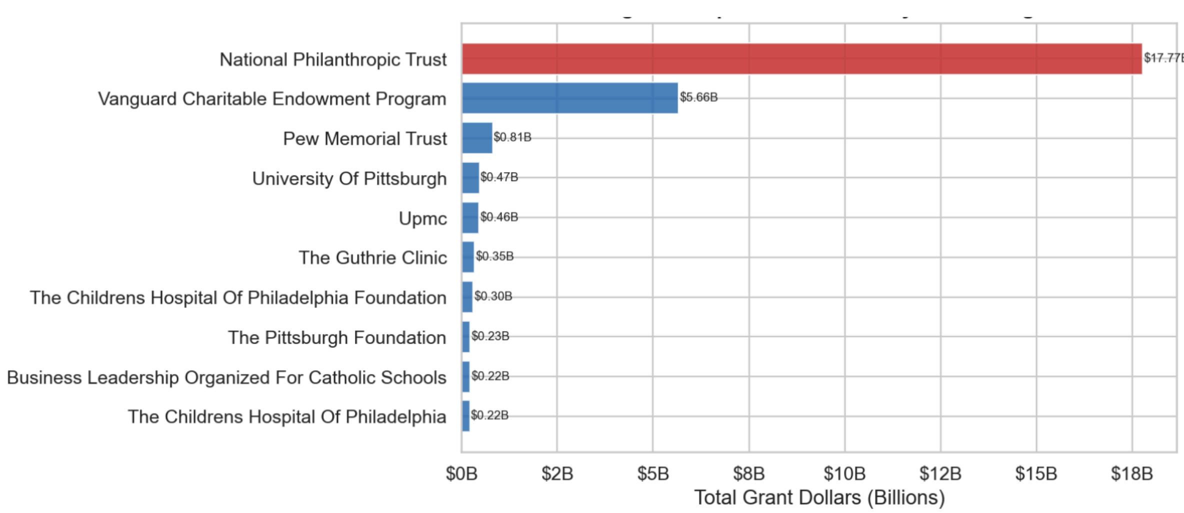

Figure 8 Top 10 Foundations by Total Giving

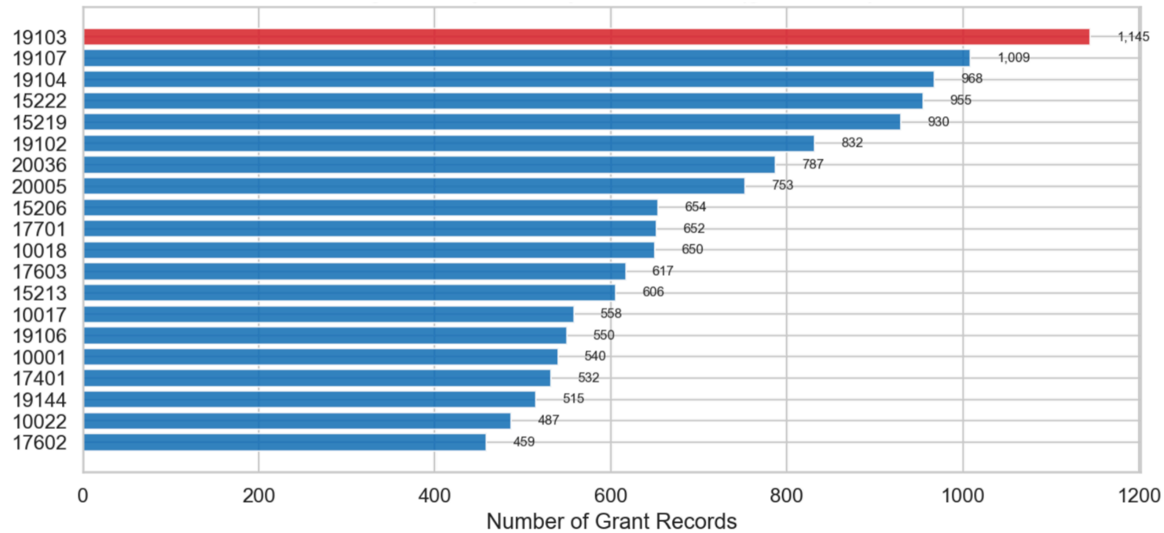

Figure 9 Top 20 recipient ZIP codes by number of grant records

Appendix B — MVP Task Templates (Copied Verbatim)

The three query templates below define the prototype's complete analytical scope. They are reproduced here as the authoritative reference for future development.

Task Template 1 — Recommend Funders for a Given Nonprofit

Trigger phrase: *"Which foundations should we approach?" / "Recommend funders for us."*
Data fields used: foundations.ntee_code, foundations.county, foundations.annual_giving, donations.grantee_ntee, donations.grantee_state
Model output: Ranked list of up to 10 foundations with: foundation name, EIN, total 2023 giving, primary cause area, geographic focus, and ≥ 1 example past grant as evidence citation. Revenue-missing flag appended where applicable.
Report section mapping: Sections 5 (EDA), 6.3, 7.1, 8.1

Task Template 2 — Find Funders by Cause Area & Geographic Proximity

Trigger phrase: *"Find funders focused on [cause] near [location]."*
Data fields used: foundations.ntee_code, foundations.county, county_lookup.latitude, county_lookup.longitude, foundations.annual_giving, foundations.revenue_flag
Model output: Proximity-sorted funder list: foundation name, distance estimate (county centroid), NTEE cause label, annual giving band, revenue-missing flag.
Report section mapping: Sections 5 (EDA), 6.2, 6.3, 7.2, 8.2

Task Template 3 — Find Similar Organizations, Then Find Their Funders

Trigger phrase: *"What foundations fund organizations like ours?"*
Data fields used: donations.grantee_ntee, donations.grantee_state, donations.grant_amount → join to foundations via donations.EIN
Model output: Two-step output: (Step 1) similar grantees (matching NTEE + geography); (Step 2) foundations that funded those grantees with grant evidence. NTEE gap flag appended when grantee code coverage < 50%.
Report section mapping: Sections 5 (EDA), 6.2, 6.3, 7.3, 8.2

Appendix C — Assumptions & Out-of-Scope List

C.1 Key Analytical Assumptions

- IRS 990 data is assumed to be materially accurate; no independent verification of reported figures is performed.
- Annual giving threshold (> $1M) is applied to the most recent available filing year (2023 where available, else 2022).
- County centroids are used as the geographic proxy for all proximity calculations; no street-level geocoding is performed.
- NTEE codes are treated as stable within a filing year; multi-year NTEE drift is not modeled.
- The 100-foundation donation sample is treated as internally consistent for prototype demonstration; representativeness claims are limited accordingly.

C.2 Out-of-Scope Items (Confirmed)

UI / Interface design	Front-end, visual design, user testing — fully out of scope
Outreach automation	Auto-drafting emails, grant proposals, or donor communication
Private data sources	Subscription databases (Foundation Directory Online, Candid Premium, etc.)
Non-PA foundations	Any foundation whose primary state of registration is not Pennsylvania
< $1M giving foundations	Foundations below the annual giving threshold are excluded from all queries
Real-time data feeds	Live API connections to IRS, Candid, or any third-party data provider
Multi-year trend modeling	Year-over-year giving trend analysis (deferred to next phase)
ML / predictive scoring	Probabilistic models of funding likelihood; this prototype uses deterministic SQL only

Appendix D — Discovery Pipeline User Guide

Document Title: Pipeline User Guide: KMSG Final Project.docx

Description:
This document provides a comprehensive, step-by-step guide to using the data pipeline developed in this project. It includes instructions for running the pipeline as-is, modifying key parameters (e.g., state, revenue thresholds, time period), and extending the system for future use cases.

The guide is written for all audiences and is intended to support the client in maintaining, reproducing, and adapting the dataset over time.

Contents include:

- Pipeline overview and logic
- Section-by-section explanation of code
- Instructions for running the pipeline
- Parameter modification guidance
- Instructions for extending to other states
- Troubleshooting and limitations

Appendix E — Ethics and Governance Memorandum

Document Title: Ethics and Governance Memorandum.docx

Description:
This memorandum establishes the ethical, policy, and governance framework for Funder Compass, an AI-assisted decision-support tool built on publicly available nonprofit grant data. It defines the system's intended use, clarifies its limitations, and outlines principles and practices necessary to ensure transparent, interpretable, and responsible use. The document also provides a structured governance approach to support ongoing maintenance, risk management, and future development.

Contents include:

- System purpose, scope, and intended use, including clarification that the tool is a decision-support system (not predictive)
- Data foundations and limitations, including reporting delays, incomplete filings, and variability in data quality
- Risk considerations, including representation bias, legacy bias, concentration effects, and over-reliance
- Built-in safeguards and governance practices, including explainability, validation, monitoring, and documentation
- Guidance for responsible use, interpretation, and future development, including user expectations and system maintenance

Appendix F — Data Discovery GitHub Repository

Document Title: KMSG Capstone — Nonprofit Funder Discovery Pipeline

Description:
The GitHub repository contains the full implementation of the project, including the end-to-end data pipeline, intermediate outputs, and documentation required to reproduce and extend the dataset.

Repository Contents:
- End-to-end pipeline code (Python)
- Intermediate datasets and outputs
- Final cleaned dataset
- README documentation explaining structure and usage

Purpose: The repository serves as the primary technical reference for the project and enables:
- Full reproducibility of results
- Transparency in data collection and processing
- Future development and extension of the pipeline

Access Repository: https://github.com/zxtine/kmsg_project.git

References & Data Access

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